NetApp Automation

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NetApp
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NetApp Automation

What’s new with NetApp Automation

NetApp regularly updates the Automation Catalog solutions and related software to bring you new features, enhancements, and bug fixes.

04 June 2023

Initial launch of the documentation site.

BlueXP automation catalog

Overview of the BlueXP automation catalog

The BlueXP automation catalog is a collection of automation solutions available to NetApp customers, partners, and employees. Some of the features and benefits of the catalog are described below.

Single location for your automation needs

You can access the automation catalog through the BlueXP web user interface. This provides a single location for the scripts, playbooks, and modules needed to enhance the automation and operation of your NetApp products and services.

Scripts are created and tested by NetApp

All the automation solutions and scripts have been created and tested by NetApp. Each solution targets a specific customer use case or request. Most focus on integration with the NetApp file and data services.

Solution documentation

Each of the automation solutions includes associated documentation to help you get started. While the solutions are accessed through the BlueXP web interface, all the documentation is available at this site. The documentation is organized based on the NetApp products and cloud services.

Solid foundation for the future

NetApp is committed to helping our customers improve and streamline the automation of their data centers and cloud environments. We expect to continue enhancing the BlueXP automation catalog to address customer requirements, technology changes, and continued product integration.

Azure Netapp Files

Install Oracle using Azure NetApp Files

You can use this automation solution to provision Azure NetApp File (ANF) volumes and install Oracle on an available virtual machine. Oracle then uses the ANF volumes for data storage.

About this solution

At a high level, the automation code provided with this solution performs the following actions:

- Set up NetApp account on Azure
- Set up storage capacity pool on Azure
- Provision Azure NetApp Volumes based on customer definition
- Create mount points
- Mount the ANF volumes to the mount points
- Install Oracle on the Linux server
- Create listeners for the database
- Create PDBs

- Starting listener and Oracle instance

The automation is based on Terraform and Ansible, both of which must be installed on the Linux virtual machine as described below.

Before you begin

You must have the following to complete the Oracle installation:

- The automation solution packaged as file `Ora_ANF_Prov.zip`. You can download the file at the BlueXP web user interface.
- A Linux VM with the following characteristics:
  - RHEL 8 (Standard_D8s_v3)
  - Public internet access
  - Deployed on the same Azure Virtual Network used for the Azure NetApp Files provisioning
- Azure account

Step 1: Install Terraform

Install Terraform which is needed to run the automation code.

Steps

1. Using the Linux command line interface, navigate to an appropriate directory for the installation.
2. Download the installation package.

```
wget
/https://releases.hashicorp.com/terraform/1.4.2/terraform_1.4.2_linux_am
d64.zip
```

3. Extract the software.

```
unzip terraform_1.4.2_linux_amd64.zip
```

4. Remove the installer file.

```
rm terraform_1.4.2_linux_amd64.zip
```

Step 2: Install Ansible and the NFS utilities

Install Ansible and the NFS utilities package which are needed to run the automation code.

Steps

1. Install the Ansible Python package.

```
sudo pip3 install ansible
```

2. Install the NFS utilities.

```
sudo yum install nfs-utils
```

Step 3: Download the Oracle installer files

Download the required Oracle installation and patch files.

Steps

1. Sign in to your Oracle account as needed.
2. Download the following files.

File	Description
LINUX.X64_193000_db_home.zip	19.3 base installer
p31281355_190000_Linux-x86-64.zip	19.8 RU patch
p6880880_190000_Linux-x86-64.zip	opatch version 12.2.0.1.23

3. Place the files in the directory `/tmp/archive` where they are needed for installation.
4. Make sure all users on the database server being deployed have read, write, and execute access to the directory `/tmp/archive`.

Step 4: Install the Oracle database

You can now install Oracle using the automation solution code.

Steps

1. Copy the solution file `Ora_ANF_Prov.zip` to the virtual machine where the automation code will be run.

```
scp -i ~/private-key.pem -r Ora_ANF_Prov.zip user@<IP_ADDRESS_OF_VM>
```

   Note the input parameter `private-key.pem` is your private key file used for Azure virtual machine authentication.

2. Unzip the solution file.

```
unzip Ora_ANF_Prov.zip
```

3. Navigate to the directory `Ora_ANF_Prov` created with the unzip operation and list the files.

```
ls -lt
```

4. Prepare the Terraform variables:
   a. Confirm the files `terraform.tfvars` and `variables.tf` exist
   b. Update `terraform.tfvars` as needed for your environment.
5. Execute the installation script in the same directory.

```
bash exec-command.sh
```

6. Sign in to Oracle as `user`.

```
sudo su oracle
```

7. Reload the Bash profile variables and confirm they are set properly.

```
cd /home/Oracle
source .bash_profile
echo $ORACLE_HOME
```

Step 5: Validate the Oracle installation

You should confirm that the installation was successful.

Steps

1. Log in to Oracle server as oracle user and display a list the Oracle processes. This will indicate if installation completed as expected and the Oracle database is running.

```
ps -ef | grep ora
```

2. Log in to the database to examine the database configuration settings and to confirm the PDBs were created properly.

```
sqlplus / as sysdba
```

   You should see output similar to the following:

```
SQL*Plus: Release 19.0.0.0.0 - Production on Thu May 6 12:52:51 2021
Version 19.8.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.8.0.0.0
```

3. Execute a few simple SQL commands to confirm the database is available.

```
select name, log_mode from v$database
show pdbs
```

NetApp product APIs

ONTAP 9

NetApp ONTAP is the industry's leading data management software for both cloud and on-premise deployments. ONTAP includes a single common REST API that continues to be expanded and enhanced with each release. See the documentation and related resources provided below to get started with the automation of your ONTAP deployments using the ONTAP REST API.

ONTAP REST API

You can use the REST API to automation the administration of your ONTAP deployments.

- ONTAP Automation documentation

ONTAP family

The ONTAP family documentation includes everything you need to install and administer your ONTAP deployments.

- ONTAP product documentation

BlueXP control plane

NetApp BlueXP BlueXP is a unified control plane that provides a hybrid multi-cloud platform for administering storage and data services across both on-premises and public cloud environments. It is composed of several distinct services or components, each of which exposes an associated REST API. See the documentation and related resources provided below to get started with the automation of your BlueXP environment using the BlueXP REST APIs.

BlueXP REST APIs

You can use the various REST APIs to automation the administration of your storage and data services managed by BlueXP.

- BlueXP API documentation

BlueXP family

The BlueXP family documentation includes everything you need to get started using the BlueXP control plane.

- BlueXP documentation

Astra Control

Astra Control is a NetApp software product that provides application-aware data management of Kubernetes clusters in multiple environments. The two deployment models share a common REST API. See the documentation and related resources provided below to get started with the automation of your Astra deployments using the

Astra Control REST API.

Astra Control REST API

You can use the REST API to automation the administration of your Astra Control Service and Astra Control Center deployments.

- Astra Control Automation documentation

Astra family

The Astra family documentation includes everything you need to install and administer the Astra and related software.

- Astra documentation

Knowledge and support

Additional resources

There are additional resources you can access to get help and find more information about NetApp products as well as the cloud services.

NetApp developer resources

- NetApp DevNet

  A central location for developer resources supporting NetApp partners and customers.

- NetApp IO - The Pub

  Blog articles and other resources to support developers and administrators.

NetApp cloud resources

- NetApp BlueXP

  Central site for the NetApp cloud solutions.

- NetApp Cloud Central console

  NetApp Cloud Central service console with sign in.

Get help

NetApp provides support for its products and cloud services in a variety of ways. Extensive free self-support options are available 24x7, such as knowledgebase (KB) articles and a community forum.

Self-support options

- Knowledge base

  Search through the knowledge base to find helpful articles to troubleshoot issues.

- Communities

  Join a NetApp community to follow ongoing discussions or create new ones.

- NetApp Support

  Access troubleshooting tools, documentation, and technical support assistance.

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